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Reducing the forage content of the ration using the undigested forage neutral detergent fiber: The effects on production, rumen environment, and digestibility

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ABSTRACT

Several factors influence the effectiveness of forage in ruminant rations, including NDF content, the physical nature, fragility, and digestibility. Recently, several studies suggested using the undigestible NDF (uNDF) fraction as a possible approach to achieve a more precise ration. The objective of the current study was to reduce the forage content of the diet by using the in vitro forage uNDF for diet formulation and to determine the effects on production, rumen environment, and digestibility. Thirty-four cows were divided into 2 groups in a crossover design study. Treatments consisted of either (1) a control (CTL) diet containing 35.8% forage (DM basis), 11.8% forage uNDF estimated through 30 h of in vitro incubation ($uNDF_{30}$), and 10% forage uNDF estimated through 48 h of in vitro incubation ($uNDF_{48}$) based on 20% wheat silage and 15.8% wheat hay, or (2) a low-forage (LF) diet containing 30.6% forage, 11.8% forage $uNDF_{30}$, and 10.4% forage $uNDF_{48}$ based on 20% wheat silage, 2.2% wheat hay, and 8.3% wheat straw. Each period lasted 35 d, and data collection occurred during the final 21 d. Milk yields were recorded daily, and milk samples were taken weekly. Two rumen samples were collected twice for VFA, pH, and ammonia, and 8 fecal samples were collected for total-tract digestibility measurements. No differences were observed in rumen pH, ammonia, and VFA concentrations. Apparent total-tract digestibility of DM, OM, protein, ether extracts, and NDF was higher in the CTL diet. Milk yields (52.2 and 51.7 kg/d, respectively), 4% FCM, and ECM yields were higher in the CTL than in the LF treatment. The milk fat and protein content did not differ, the milk fat yield tended to be higher, and the milk protein yield was higher in the CTL treatment. The DMI was 3.2% higher in the CTL than in the LF treatment (32.7 vs. 31.7 kg/d, respectively), and the milk-to-DMI ratio was higher in the LF than in the CTL treatment. In summary, reducing the forage content by balancing the diet for forage uNDF reduced the DMI, milk, fat, and protein yields and digestibility, probably due to the high inclusion rate of wheat straw in the LF diet. However, the production efficiency of milk was higher in the LF treatment. Further research is required to fine-tune the proportion of forage uNDF and to determine the optimal exchange of forage feedstuffs in the diet.
Key words: forage, NDF digestibility, rumen pH, uNDF, wheat straw

INTRODUCTION

Adequate dietary fiber is important for dairy cattle to maintain rumen health and to optimize ruminal microbial efficiency (Eastridge et al., 2009). Moreover, assessing the effectiveness of ration to maintain proper rumen function and overall health and to maximize production is one of the most challenging aspects of balancing the diet for high-producing dairy cows. Traditional parameters, such as forage and NDF content in the ration, do not consider the physical characteristics of the forage and its degradability in the rumen (White et al., 2017). The term "physical effective NDF" (**peNDF**) was first introduced by Mertens (1997). Although it includes the particle size distribution in the ration, it was not included in the NRC 2001 model, which provided guidelines that include forage NDF, total NDF, and NFC content in the ration (NRC, 2001). NASEM (2021) adopted the model suggested by White et al. (2017), which uses the term "physically adjusted NDF" (**paNDF**). This model uses factors such as the total forage, NDF, forage NDF, and starch, as well as the fraction of the 19-mm and the 8-mm sieve in the Penn State Particle Separator (**PSPS**) of the TMR to evaluate

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the ration characteristics required to maintain >6.0 rumen pH. However, this model does not consider the variability in NDF degradation rate and its effect on the rumen. Furthermore, both peNDF and paNDF are affected by time and the mixing processes, which makes them less effective for balancing diets when using forage such as hay and straw that are not prechopped.

Although research over the past 60 years has focused on the digestible fraction of the NDF, in recent years, there has been emerging research that tested the direct effect of the undigested fraction of the NDF (**uNDF**) on feed intake, rumen function, and milk production (Cotanch et al., 2014; Fustini et al., 2017; Grant et al., 2021). In a study conducted in northern Italy that examined rations with 2 different concentrations of roughage and 2 sources of alfalfa hay that differed in their NDF digestibility (**NDFd**), it was found that the consumption of uNDF estimated through 24 h of in vitro incubation (**$uNDF_{24}$**) was constant at 0.74% of the cow's BW. No difference was found in the milk yield; however, the milk fat content was higher in cows fed the lower digestible alfalfa (Fustini et al., 2017). In another study, wheat straw replaced corn silage or alfalfa hay, while maintaining a constant concentration of forage uNDF estimated through 30 h of in vitro incubation (**$uNDF_{30}$**) and the NDFd estimated through 30 h of in vitro incubation (**$NDFd_{30}$**) in the diets (Kahyani et al., 2019b). It was found that in a ration in which alfalfa hay was replaced with wheat straw (while lowering the concentration of roughage in the ration from 40% to 31%), no differences were observed in DMI, milk yield, rumen pH, and digestibility (Kahyani et al., 2019b). These studies suggest that using the nondigestible fraction of NDF might be a possible approach to achieve a more precise ration, while lowering the total forage content and maintaining high milk production without harming the cow's health.

One limitation in diet formulation using the uNDF parameter is that uNDF alone is apparently insufficient to determine the effectiveness of the roughage in the ration. Grant et al. (2020) examined the relationship between uNDF estimated through 240 h of in vitro incubation (**$uNDF_{240}$**) and the concentration of peNDF using 2 levels of the forage-to-concentrate ratio and 2 levels of hay chop length. Interestingly, they found that chopping hay fine (24% > 1.18 mm) in a ration high in $uNDF_{240}$ achieved a DMI similar to that of a ration with a low concentration of $uNDF_{240}$ but with a coarser cut (58% > 1.18 mm).

Currently, very few publications have evaluated the effects of using uNDF as a factor in diet formulation. Generally, in the field, the uNDF content is determined only for forage rather than for all ingredients, and therefore, in this study, we used the forage uNDF factor. The objectives of the current study were to use the forage uNDF factor in diet formulation while reducing the forage content by substituting wheat sources between diets, and to examine the effects on production, rumen environment, and digestibility. Our hypothesis was that balancing the diet for forage uNDF would have a similar effect on feed intake and production and help to reduce the forage content of diets while maintaining proper rumen function and cow performance.

MATERIALS AND METHODS

Animals and Experimental Design

The experimental protocol for the study was approved by the Volcani Center Animal Care Committee (approval number 921/21 IL) and was performed in accordance with the relevant guidelines and regulations. The experiment was conducted at the Volcani Center experimental farm in Rishon LeZion, Israel. Thirty-four multiparous high-yielding Israeli-Holstein dairy cows were housed in a covered loose pen with adjacent outside yards that were equipped with a real-time electronic individual feeding system. Each feeding station was equipped with an individual identification system (ID tag, S.A.E. Kibbutz Afikim, Israel) that allowed each cow to enter a specific feeding station and record the daily feed intake. After 10 d of adapting to the feeding stations, the cows were divided into 2 treatment groups, each consisting of 17 cows. Cows were divided by milk production, DIM, parity, and BW, and then randomly assigned to one of the 2 treatment groups. In a previous study, we determined the uNDF fraction after 30, 48 (**$uNDF_{48}$**), and 240 h of incubation of all the ingredients used in the current study. The experimental diets were balanced for starch, CP, and forage $uNDF_{30}$ and forage $uNDF_{48}$ content. The cows were fed a milking-cow ration that contained either (1) a control (**CTL**) diet consisting of 35.8% forage (DM basis) based on 20% wheat silage, 15.8% wheat hay, 11.8% forage $uNDF_{30}$, and 10.0% forage $uNDF_{48}$ or (2) a low-forage (**LF**) diet consisting of 30.6% forage based on 20% wheat silage, 2.2% wheat hay, 8.3% wheat straw, 11.8% forage $uNDF_{30,}$ and 10.4% forage $uNDF_{48}$. The ration's composition and content are presented in Table 1. The mean ± SD of all cows at the start of the nutritional treatments was as follows: milk yield, 55.6 ± 4.9; parity number, 3.4 ± 1.5; DIM, 137 ± 58; and BW 663 ± 48 kg. The TMR was prepared daily using a self-mixer wagon (Siloking, Tittmoning, Germany). A constant mixing time was maintained, and a premix of all the concentrate feedstuff, vitamins, and minerals was used to reduce the day-to-day variance in diet composition. Cows were fed once a day at 1000 h with 105% of the expected intake, which was adjusted daily according to the previous day's intake. The chemical composition and

Table 1. Ingredients of the experimental diets

Item	Diet[1] CTL	LF
Wheat silage	20.0	20.0
Wheat hay	15.7	2.2
Wheat straw	0.0	8.3
Corn grain ground	18.5	18.5
Barley grain rolled	3.1	3.1
Wheat grain ground	2.2	4.0
Soybean meal	5.0	5.0
Canola meal	4.5	4.5
Wheat bran	3.3	6.6
Corn gluten feed	12.4	12.4
DDGs[2]	6.6	6.6
Cotton seed	1.9	1.9
By-product of dairy industry	2.3	2.3
Urea	0.1	0.2
Calcium soaps of fatty acids	2.0	2.1
Oil	0.1	0.1
Salt:$CaCO_3$[3]	1.4	1.3
Limestone	0.2	0.1
Sodium bicarbonate	0.7	0.7
Vitamins premix[4]	0.1	0.1

[1]CTL = 35.7% forage (DM basis), 11.8% forage $uNDF_{30}$ and 10.0% forage $uNDF_{48}$ or LF = 30.5% forage, 11.8% forage $uNDF_{30}$, and 10.4% forage $uNDF_{48}$.

[2]Corn dried distillers grains with solubles.

[3]Calcium carbonate and salt mixture in a 70:30 ratio.

[4]Contained 20,000,000 IU/kg of vitamin A; 2,000,000 IU/kg of vitamin D; 15,000 IU/kg of vitamin E; 6,000 mg/kg of Mn; 6,000 mg/kg of Zn; 2,000 mg/kg of Fe; 1,500 mg/kg of Cu; 120 mg/kg of I; 50 mg/kg of Se; and 20 mg/kg of Co.

particle size distribution of the forage and the TMR are presented in Table 2. The study continued for 10 wk in a crossover design. Each study period began with 2 wk of adaptation, followed by 3 wk of data collection. Cows were milked 3 times daily, and yields were recorded automatically (AfiLab, Afimilk, Kibbutz Afikim, Israel). Cows were weighed 3 times a day after each milking using a walking electronic scale (Afiweight scale, Afimilk, Kibbutz Afikim, Israel). During the entire study, health events were monitored and recorded. Milk samples were collected from 3 consecutive milkings every 7 d during the collection period (in total, 6 milk tests for each cow) and analyzed for milk fat, protein, lactose, and urea using infrared (standard IDF 141C:2000; IDF, 2000) at the laboratories of the Israeli Cattle Breeders' Association (Caesarea, Israel). The SCC was also determined by the same laboratory. Rumination, eating, and panting time were measured using the HR-LDn automated monitoring device (SCR Engineers Ltd., Netanya, Israel). Data Flow software (version 21.1.8.0, SCR Engineers Ltd., Netanya, Israel) was used to analyze the rumination, eating, and total chewing time as minutes within a day at 2-min resolution (Merenda et al., 2019). The experiment was carried out from May to July, which is part of the hot season in Israel. Thus, both groups were exposed daily to 5 cooling sessions: 3 before each milking and another 2 sessions between milkings. Each cooling session lasted 45 min; it consisted of repeated cycles of 30 s of showering and 4.5 min of forced ventilation without showering.

Particle Size and Sorting Behavior

Twice a week during the last 21 d of each period, samples of TMR and orts were collected for particle size measurement. Particle size was measured in triplicate using the PSPS equipped with 3 sieves and a bottom pan (Kononoff et al., 2003) with the modification of the lower pen to 4 mm instead of 1.18 mm as suggested by Maulfair et al.,(2011). During wk 5 of each collection period, 12 cows from each group were subjected to a feed sorting test. Sorting of particles was determined from the actual intake of each fraction compared with the predicted intake of the same fraction had the diet been consumed as formulated (Leonardi and Armentano, 2003). Due to the unique feeding behavior of cows eating from individual feeding bins, which prevents cows from sorting when the feeding bin is almost empty, we decided to perform the feed sorting test 10 h after feed delivery, and not as traditionally done after 24 h, in which the cows had not yet eaten most of the TMR, as in Shaani et al. (2017). A sorting index score was calculated as the percentage of actual intake compared with the predicted intake for each fraction of the PSPS. Values equal to 100% indicated no sorting, whereas values <100% indicated selective refusals (sorting against), and values >100% indicated preferential consumption (sorting for).

Total-Tract Apparent Digestibility

At wk 4 of each period, 12 cows were randomly selected from each group for apparent total-tract digestibility measurements. Fecal samples were collected by rectal palpation at 6-h intervals over a 48-h period, for a total of 8 fecal samples per cow in 3-h intervals. The fecal samples were then air-dried to a constant weight, at 60°C for 7 d. The dried fecal samples were coarsely ground and combined on an equal DM basis into one representative fecal sample per cow. Next, the dried samples were analyzed for DM, NDF, CP, starch, ash, and fat. The indigestible NDF (**iNDF**) was used as an internal marker (Morris et al., 2018). To determine the iNDF concentration for each cow, 3 replicates from the dried and ground fecal samples were sealed in an F57 filter bag (Ankom Technology, Macedon, NY) and incubated in the Daisy[II] incubator (Ankom Technology, Macedon, NY) for 240 h, as described by Valentine et al. (2019). In addition, 0.5-g aliquots were taken from the diet samples, put into F57 filter bags, in 5 replicates for each diet, and placed in the Daisy[II] incubator for

Table 2. Chemical composition of the forage and the experimental TMR (mean ± SD)[1]

Item[3]	Forage: Wheat hay	Forage: Wheat straw	Forage: Wheat silage	Diet[2]: CTL	Diet[2]: LF
Chemical composition, % DM					
DM	89.3 ± 0.1	91.7 ± 0.1	36.5 ± 0.3	66.0 ± 1.8	66.0 ± 2.4
OM	89.1 ± 0.7	91 ± 0.8	90.3 ± 0.6	91.3 ± 0.2	91.5 ± 0.4
CP	8.2 ± 0.1	3.6 ± 0.1	8.5 ± 0.1	16.1 ± 0.2	16.0 ± 0.1
EE	2.7 ± 0.1	2.3 ± 0.1	4.1 ± 0.1	5.2 ± 0.1	5.3 ± 0.5
NDF	56.8 ± 0.4	78.0 ± 0.4	49.8 ± 0.3	34.5 ± 0.9	36.7 ± 1.3
ADF	33.4 ± 0.2	49.3 ± 0.2	30.8 ± 0.2	17.3 ± 0.7	18.3 ± 0.9
NFC	21.4 ± 0.4	7.2 ± 0.2	27.8 ± 0.4	39.0 ± 0.4	37.4 ± 1.2
Starch	6.9 ± 0.5	0.6 ± 0.1	10.7 ± 0.3	25.0 ± 1.1	24.8 ± 3.2
Forage	—	—	—	35.7	30.5
NEL, Mcal/kg of DM				1.73	1.73
Forage NDF	—	—	—	18.9[a] ± 1.2	17.7[b] ± 1.3
$NDFd_{30}$, % NDF	41.2 ± 0.7	30.7 ± 1.4	34.2 ± 0.9	38.5 ± 1.5	36.6 ± 1.8
$uNDF_{30}$, % DM	33.4 ± 0.6	54.1 ± 1.1	32.8 ± 0.7	21.2[b] ± 0.2	23.3[a] ± 0.3
Forage $uNDF_{30}$, % DM	—	—	—	11.8 ± 0.3	11.8 ± 0.3
$NDFd_{48}$, % NDF	53.3 ± 0.5	37.7 ± 0.4	41.9 ± 0.6	48.5 ± 1.3	46.5 ± 1.7
$uNDF_{48}$, % DM	26.6 ± 0.3	48.6 ± 0.4	29.0 ± 0.2	17.0[b] ± 0.2	19.7[a] ± 0.3
Forage $uNDF_{48}$, % DM	—	—	—	10.0 ± 0.2	10.4 ± 0.3
$NDFd_{240}$, % NDF	64.0 ± 0.3	66.1 ± 0.3	68.9 ± 0.4	69.6[a] ± 1.0	65.8[b] ± 1.3
$uNDF_{240}$, % DM	20.4 ± 0.3	26.5 ± 0.1	15.5 ± 0.4	10.5[b] ± 0.1	12.6[a] ± 0.2
Forage $uNDF_{240}$, % DM	—	—	—	6.3[a] ± 0.6	5.7[b] ± 0.8
$peNDF_{4mm}$, % of DM	—	—	—	15.3 ± 1.3	16.7 ± 1.4

[a,b]Within diets (CTL vs. LF) different superscript letters are statistically different ($P < 0.05$).

[1]Standard deviation of diets regarding forage NDF, forage NDFd, and forage uNDF was calculated based on the forage NDF and NDF digestibility assays on the day of sampling.

[2]CTL = 35.7% forage (DM basis) based on 20% wheat silage, 15.8% wheat hay, 11.8% forage $uNDF_{30}$, and 10.0% forage $uNDF_{48}$ or LF = 30.5% forage based on 20% wheat silage, 2.2% wheat hay, 8.3% wheat straw and 10.4% forage $uNDF_{48}$.

[3]NDF = NDF measured using amylase and sodium sulfite; NEL = net energy calculated according to the NASEM (2021) energy model; $NDFd_{30}$ = NDF digestibility estimated through 30 h of in vitro incubation; $uNDFd_{30}$ = uNDF estimated through 30 h of in vitro incubation; $NDFd_{48}$ = NDF digestibility estimated through 48 h of in vitro incubation; $uNDFd_{48}$ = uNDF estimated through 48 h of in vitro incubation; $NDFd_{240}$ = NDF digestibility estimated through 240 h of in vitro incubation; $uNDFd_{240}$ = uNDF estimated through 240 h of in vitro incubation.

240 h to determine the iNDF content. After the fecal and dietary samples were removed, they were air-dried at 60°C for 48 h. The iNDF content was measured and used as an internal marker for the apparent total-tract digestibility calculation. The digestibilities of DM, OM, NDF, CP, starch, and fat were determined based on the following equations (Moallem et al., 2009):

$$\text{Digestibility of DM} = 100 - \left[100 \times \left(\frac{\%\ \text{iNDF in diet}}{\%\ \text{iNDF in feces}}\right)\right],$$

$$\text{Digestibility of nutrient} = 100 - \left[100 \times \left(\frac{\%\ \text{iNDF in diet}}{\%\ \text{iNDF in feces}}\right) \times \left(\frac{\text{Nutrient in feces}}{\text{Nutrient in diet}}\right)\right].$$

NDF In Vitro Digestibility and uNDF

We determined the NDFd and uNDF content for all forage sources and TMR according to the method described by Alende et al. (2018). Briefly, 2 ruminal cannulated nonlactating Holstein cows were fed a high-forage TMR diet (35% wheat hay, 13% wheat silage, 25% wheat straw, and 27% concentrate mix; DM basis) as recommended by Krizsan and Huhtanen (2013). Dried samples were ground to pass the 1-mm screen of a Wiley mill (Arthur H. Thomas, Philadelphia, PA), and 0.5 g of each sample was weighed and placed into an Ankom F57 bag with a pore size of 25 µm. Samples were incubated using the $Daisy^{II}$ incubator for 30, 48, and 240 h in triplicate, along with 3 blank bags for each time point. After their removal, samples were soaked in cold water and then washed in cold water to remove any digesta residue. The $uNDF_{30}$, $uNDF_{48}$, and $uNDF_{240}$ residues were the NDF residues after 30-, 48-, and 240-h incubations, respectively. The NDFd value at each time point was calculated as

$$\text{NDFd (\% of NDF)} = 100 \times (\text{initial NDF} - \text{NDF residue})/\text{initial NDF}.$$

Rumen Sampling and Analysis

In the second week of each collection period, 12 cows from each group were randomly selected and sampled for rumen fluid collection for analysis of ammonia and VFA concentrations. Rumen fluid samples were collected using a stomach tube, 2 and 5 h post-feeding. Rumen samples were collected by trained personnel, and precautions were taken to avoid contamination with saliva: a vacuum pump was turned on only after ensuring that the tube was positioned inside the rumen; in addition, the first 250 mL obtained was discarded. Filtered rumen fluid (5 mL) was immediately mixed with 20% trichloroacetic acid (1:1 vol/vol) and frozen at −20°C. These rumen samples were later used to determine the ammonia concentration using the phenol procedure (Chaney and Marbach, 1962). Next, 700 μL of 10% $HgCl_2$ solution was added to another 10 mL of filtered ruminal fluid, which was then centrifuged at 3,000 × g at 15°C for 10 min, before being stored at −20°C. The VFA in the centrifuged ruminal fluid samples were determined using a 5890 series 2 gas chromatograph (Agilent Technologies, Wilmington, DE) equipped with a capillary column (30 m × 0.53 mm, 0.5 mm i.d.; Agilent Technologies, Santa Clara, CA) and a flame ionization detector. The injection port, column, and detector were maintained at 175°C, 130°C, and 165°C, respectively.

In addition, 7 cows from each treatment group were equipped with an indwelling wireless Moonsyst bolus (MOON; International, Kinsale, Republic of Ireland). This bolus was recently compared with other commercial boluses in a dairy cattle experiment with rumen-cannulated heifers (Weir et al., 2024). The units (3.2 cm i.d., 13 cm long, and weighing 180 g) were manually inserted into the reticulorumen via the esophagus on d 0 of the first period. The boluses' pH and temperature were measured every 10 min. The recorded data were automatically transmitted via wireless technology from a receiver in the barn. The measurements were taken continuously from each cow until the end of the experiment. The pH data from d 15 to 35 of each period were analyzed to determine the hourly and daily mean pH, and time (min/d) below specific cutoff points (5.8 and 6.0).

Statistical Analysis

Continuous variables (milk, milk solids, DMI, and efficiency calculations) were analyzed as repeated measurements using the MIXED Procedure of SAS (version 9.2, 2008; SAS Institute, Cary, NC), according to the following model:

$$Y_{ijkl} = \mu + Ti + Pj + B_k + C_l(T \times P)_{ij} + \alpha \times D_m + \beta \times R_n + \gamma_i \times R_n + E_{ijklmn},$$

where μ = the overall mean, Ti = the fixed effect of treatment (i = 1 to 2), Pj = the fixed effect of parity (j = 2 or >2), B_k = the fixed effect of period (k = 1 or 2), C(T × P)($l \times i \times j$) = the random effect of cow_l nested within $treatment_i$ and $parity_j$, D_m = DIM, α = linear coefficient of DIM, R_n = time in the study (d), β = linear coefficient of time, γi = addition to β for treatment i, and E_{ijklmn} = the residual error.

The production data, DMI, and efficiency calculation data were analyzed with the relevant specific data of the pretreatment period as covariates. The interactions between time and treatment, parity and treatment, and between DIM and treatment were included in the model; they were found to be insignificant ($P > 0.20$) and, therefore, were excluded from the model.

The autoregressive order 1 was used as a covariance structure in the model because it resulted in the lowest Bayesian information criterion for most of the tested variables.

Rumen measurements (ammonia and VFA concentrations) were analyzed as repeated measurements using the General Linear Models procedure of SAS (version 9.2, 2008; SAS Institute, Cary, NC). The model included the effect of treatment (CTL or LF), and the time of sampling (2 or 5 h post-feeding). Nutrient digestibility was analyzed using the General Linear Models procedure of SAS (version 9.2, 2002; SAS Institute, Cary, NC). The LSM and the SEM are presented in the relevant tables. Significance was declared at $P \leq 0.05$, and tendencies were reported at $0.05 < P \leq 0.10$, unless otherwise stated.

RESULTS

Forage and Diets

Chemical composition of the forage ingredients and rations, as well as the in vitro NDFd, are presented in Table 2. Average NDF content of the wheat hay was 56.8%, and in the wheat straw it was 78.0% (DM basis). The uNDF at 48 h of incubation was 1.8-fold higher in wheat straw than in wheat hay (48.6% and 26.6%, respectively, $P < 0.001$). However, no differences were observed between wheat hay and wheat straw regarding $uNDF_{240}$ ($P = 0.3$). The diets were balanced for forage $uNDF_{30}$ and forage $uNDF_{48}$ using the prestudy measurements. The actual diet $uNDF_{30}$ ($P = 0.04$), $uNDF_{48}$ ($P = 0.02$), and $uNDF_{240}$ ($P < 0.001$) were higher in the LF than in the CTL diet. The actual forage $uNDF_{48}$ was 10.0% and 10.4% in the CTL and LF diets, respectively ($P = 0.09$), and the forage $uNDF_{30}$ were similar between diets (11.8%; $P = 0.93$); however, forage $uNDF_{240}$ was lower in the LF than in the CTL diet (5.7% and 6.3%, respectively; $P = 0.03$).

Particle Size and Feed Sorting

The particle size distribution of the TMR differed for the >4- to <8-mm fractions ($P = 0.002$), but not for the other fractions (Table 3). The feed sorting test was carried

Table 3. Particle size distribution and sorting index of TMR for the different treatments

	Treatment[1]			*P*-value	
Parameter	CTL	LF	SEM	Treatment	Period
Particle size, %					
>19.0 mm	19.9	21.5	0.73	0.13	<0.001
>8.0 and < 19.0 mm	13.2	13.8	0.42	0.35	<0.001
>4.0 and < 8.0 mm	11.2	10.0	0.24	0.002	0.47
<4 mm	55.6	54.7	0.82	0.42	0.09
Sorting index,[2] %					
Intake, % of feed delivered	62.8	66.1	2.1	0.35	0.99
>19.0 mm	90.9	92.7	2.39	0.13	0.002
>8.0 and < 19.0 mm	87.3	99.0	3.14	0.01	0.13
>4.0 and < 8.0 mm	103.0	93.0	2.01	0.001	0.76
<4 mm	106.8	105.4	1.27	0.44	0.06

[1]CTL = 35.7% forage (DM basis) based on 20% wheat silage, 15.8% wheat hay, 11.8% forage $uNDF_{30}$, and 10.0% forage $uNDF_{48}$ or LF = 30.5% forage based on 20% wheat silage, 2.2% wheat hay, 8.3% wheat straw 11.8% forage $uNDF_{30}$, and 10.4% forage $uNDF_{48}$.

[2]Sorting index was measured 10 h after feed delivery as described by (Leonardi and Armentano, 2003).

out 10 h after feed delivery. We found feed sorting against long particles (<19 mm) in both groups. Feed sorting for particles >8.0 and <19.0 mm was higher for the LF treatment ($P = 0.01$); in the CTL treatment, there was sorting for >4- and <8-mm particles, whereas in the LF treatment, there was sorting against this fraction ($P = 0.001$). In addition, in both groups, the cows sorted for particles <4 mm.

Rumen pH, VFA, Ammonia, and Apparent Total-Tract Digestibility

Rumen pH, as obtained from the indwelling boluses, is presented in Table 4 and Figure 1. Although the overall average rumen pH did not differ between the treatments, the rumen pH differed at specific hours throughout the day. The mean rumen pH when consuming the CTL diet was higher during the afternoon hours (1200, 1300, 1400, and 1500 h) and evening hours (1800, 1900, 2000, 2100, and 2300 h) than when consuming the LF diet ($P < 0.05$). These changes resulted in a lower minimum daily pH ($P = 0.02$) and extended for a longer time during the day with pH less than 6.0 ($P = 0.004$) and 5.8 ($P = 0.01$). In addition, the mean rumen pH and minimum and maximum daily rumen pH were higher in the second period than in the first period ($P = 0.01$). There was no interaction between treat-

Table 4. Least squares means of rumen pH, and ammonia and VFA concentration

	Treatment[1]			*P*-value[2]	
Parameter	CTL	LF	SEM	Treatment	Period
pH[2]					
Mean pH	6.36	6.31	0.038	0.15	<0.01
pH <6.0, min/d	52.7	171.0	40.6	0.004	0.08
pH <5.8, min/d	26.8	110.9	32.6	0.01	0.05
Minimum pH	5.99	5.86	0.053	0.02	<0.01
Maximum pH	6.79	6.80	0.031	0.83	<0.01
VFA[3]					
Acetate, mM	59.4	59.8	1.12	0.78	0.31
Propionate, mM	29.3	30.4	0.62	0.22	0.007
Butyrate, mM	15.1	14.9	0.47	0.85	0.14
Isovalerate, mM	1.0	1.1	0.04	0.04	<0.001
Valerate, mM	1.6	1.6	0.07	0.62	<0.001
Caproate, mM	0.4	0.5	0.03	0.11	0.36
Total VFA, mM	106.7	108.3	1.88	0.54	0.03
Acetic:Propionic	2.05	1.99	0.035	0.25	0.04
NH_3^+, mg/L	94.4	106.2	7.69	0.25	0.94

[1]CTL = 35.7% forage (DM basis) based on 20% wheat silage, 15.8% wheat hay, 11.8% forage $uNDF_{30}$, and 10.0% forage $uNDF_{48}$ or LF = 30.5% forage based on 20% wheat silage, 2.2% wheat hay, 8.3% wheat straw 11.8% forage $uNDF_{30}$, and 10.4% forage $uNDF_{48}$.

[2]pH was measured by indwelling boluses installed in 7 cows in each group.

[3]VFA was measured twice, at 2 and 5 h after feed delivery.

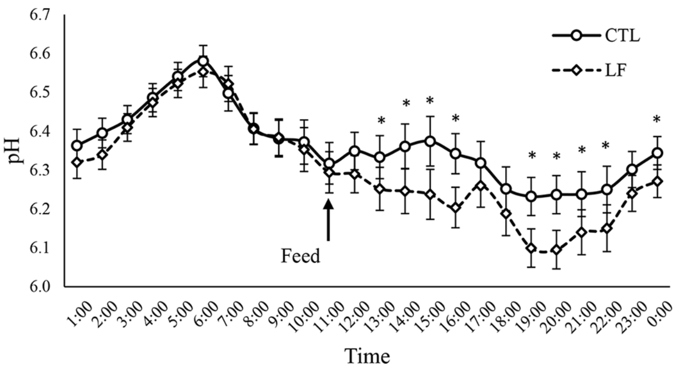

Figure 1. Rumen pH dynamics throughout the day in cows fed CTL or LF diets. The rumen pH was measured by indwelling boluses installed in 7 cows from each group, and was recorded every 10 min. Treatments: CTL = 35.7% forage (DM basis) based on 20% wheat silage, 15.8% wheat hay, and 10.0% forage $uNDF_{48}$, or LF = 30.5% forage based on 20% wheat silage, 2.2% wheat hay, and 8.3% wheat straw and 10.4% forage $uNDF_{48}$. The arrow denotes the time of day at which the cows were fed. Error bars represent the SEM. *Means between treatments differ ($P < 0.05$).

ment and period for all outcomes. The average ambient temperature was 24.3°C and 26.6°C ($P < 0.001$), and the average relative humidity was 66.3 and 66.5 during the first and the second study periods ($P = 0.60$), respectively. The calculated temperature-humidity index (**THI**; Bohmanova et al., 2007) was 3.7 units higher during the second period compared with the first period (75.3 and 71.9, respectively; $P < 0.001$). We did not find any differences in the ammonia and VFA concentration in the rumen between treatments. However, the rumen propionate, valerate, and isovalerate concentrations were affected by the period.

Apparent total-tract digestibilities of DM and OM were higher for the CTL than for the LF diet; 5.8 and 5.5 percentage units, respectively ($P < 0.001$; Table 5). In addition, the digestibilities of starch ($P = 0.05$), ether extract (**EE**; $P = 0.001$), and NDF ($P < 0.001$) were higher for cows fed the CTL diet. No differences were observed for the total-tract digestibility of CP and NFC. The effect of the period was significant for the digestibility of DM, OM, NFC, EE, and NDF.

Milk and Milk Solid Yields, Feed Intake, and Efficiency

The milk yield was 0.5 kg/d higher for cows fed the CTL diet ($P < 0.009$; Table 6 and Figure 2a). Milk fat, protein, and lactose contents did not differ between groups; however, the milk protein yield was 2.4% higher ($P = 0.03$), and the milk fat ($P = 0.06$) and lactose ($P = 0.08$) yields tended to be higher in cows fed the CTL diet rather than the LF diet. The MUN content tended to be higher in the CTL treatment ($P = 0.10$), and 4%

Table 5. Total-tract apparent digestibility of diets and nutrients

Parameter	Treatment[1] CTL	LF	SEM	P-value Treatment	Period
DM	62.6	56.8	0.56	<0.001	<0.001
OM	65.1	59.6	0.52	<0.001	<0.001
CP	61.9	53.8	1.03	<0.001	0.53
Starch	97.6	95.0	0.92	0.05	0.11
NFC	92.3	90.9	0.76	0.24	<0.001
EE	78.4	73.3	1.06	0.001	0.003
NDF[2]	36.8	30.8	1.09	<0.001	<0.001

[1]CTL = 35.7% forage (DM basis) based on 20% wheat silage, 15.8% wheat hay, 11.8% forage $uNDF_{30}$, and 10.0% forage $uNDF_{48}$ or LF = 30.5% forage based on 20% wheat silage, 2.2% wheat hay, 8.3% wheat straw 11.8% forage $uNDF_{30}$, and 10.4% forage $uNDF_{48}$.

[2]NDF = NDF measured using amylase and sodium sulfite.

FCM ($P = 0.05$; Figure 2b) and ECM ($P = 0.01$) yields were higher in cows fed the CTL diet. Milk production efficiency was 3.1% higher for cows fed the LF diet ($P < 0.001$), with no differences for 4% FCM and ECM. The changes in BW during each period were minimal (less than 3 kg/21 d) and did not differ between the dietary treatments. The period effect was significant for milk yield, milk fat, and protein content, as well as protein and lactose yield. In addition, the effect of period was significant for 4% FCM and ECM production efficiency.

The DMI was lower for the LF compared with the CTL treatment by 1.0 kg/d ($P < 0.001$), and it was also affected by period (Table 7 and Figure 2c). However, the consumption of $uNDF_{30}$ ($P < 0.001$), $uNDF_{48}$ ($P = 0.002$), and $uNDF_{240}$ ($P < 0.001$) was higher in cows fed the LF diet than in those fed the CTL diet. Consumption of forage $uNDF_{30}$ was higher ($P < 0.001$) in the CTL, and the consumption of the forage $uNDF_{48}$ was similar between treatments. The forage $uNDF_{240}$ was higher in the CTL treatment than in the LF treatment ($P = 0.001$). No differences between groups were found in BW and metabolic BW (**$BW^{0.75}$**). We also calculated the intake of nutrients per $BW^{0.75}$ (Table 7). The intake of $uNDF_{30}$ ($P < 0.001$), $uNDF_{48}$ ($P = 0.003$), and $uNDF_{240}$ ($P = 0.04$) per $BW^{0.75}$ was higher for the LF treatment than for the CTL treatment. However, the forage $uNDF_{30}$ ($P < 0.001$) and $uNDF_{240}$ intake were higher in the CTL treatment than in the LF treatment ($P < 0.001$). Remarkably, forage $uNDF_{48}$ intake per $BW^{0.75}$ did not differ between groups ($P = 0.47$). However, in addition, the forage $uNDF_{30}$ ($P < 0.001$) and the forage $uNDF_{240}$ ($P = 0.002$) intake were affected by period.

Recumbence, Rumination, Eating, and Panting Times

The dietary treatment did not affect the daily recumbence and panting times (Table 8). However, the daily eating time

Table 6. Least squares means of yields, production efficiency, and BW by treatment group

Parameter	Treatment[1]		SEM	P-value	
	CTL	LF		Treatment	Period
Milk, kg/d	52.2	51.7	0.64	0.009	0.005
Fat, %	3.82	3.73	0.071	0.15	<0.001
Protein, %	3.19	3.18	0.032	0.21	<0.001
Lactose, %	4.95	4.95	0.023	0.89	0.35
Fat, kg/d	1.99	1.91	0.034	0.06	0.78
Protein, kg/d	1.67	1.63	0.023	0.03	0.002
Lactose, kg/d	2.61	2.56	0.038	0.08	<0.001
MUN, mg/dL	15.5	14.8	0.345	0.10	0.06
SCS[2]	2.39	2.38	0.247	0.94	0.002
4% FCM, kg/d	50.5	49.5	0.51	0.05	0.58
ECM, kg/d	50.8	49.8	0.51	0.01	0.68
Milk energy, Mcal/d	37.6	36.9	0.38	0.01	0.71
Milk/DMI, kg	1.61	1.66	0.043	<0.001	0.20
4% FCM/DMI	1.57	1.59	0.041	0.24	0.002
ECM/DMI	1.58	1.60	0.039	0.14	<0.001
BW, kg	666	662	9.1	0.77	0.81
BW change, kg/period	2.9	1.9	1.28	0.12	0.57

[1]CTL = 35.7% forage (DM basis) based on 20% wheat silage, 15.8% wheat hay, 11.8% forage uNDF$_{30}$, and 10.0% forage uNDF$_{48}$ or LF = 30.5% forage based on 20% wheat silage, 2.2% wheat hay, 8.3% wheat straw 11.8% forage uNDF$_{30}$, and 10.4% forage uNDF$_{48}$.

[2]SCS = $-\text{Log}_2$(SCC/100,000) + 3.

was higher for cows fed the LF diet ($P < 0.001$). These differences were maintained both for eating time per DMI and NDF intake ($P < 0.001$), but not for eating time per uNDF$_{30}$ and uNDF$_{48}$. All parameters of eating time per NDF, uNDF, and forage uNDF were affected by period. The daily rumination time was significantly higher for cows fed the CTL diet ($P = 0.001$), as was the rumination time per kilogram of uNDF$_{30}$, uNDF$_{48}$ intake, and uNDF$_{240}$ ($P < 0.001$). In addition, the rumination times per DMI and per kilogram of NDF consumed were affected by period. Remarkably, the daily rumination time per forage uNDF$_{30}$ and uNDF$_{48}$ did not differ between treatments. The total chewing time and the total chewing time per forage uNDF$_{30}$ and forage uNDF$_{48}$ intake were not different between the dietary treatments. However, the total chewing time per uNDF$_{48}$, uNDF$_{240,}$ and forage uNDF$_{240}$ was higher for cows fed the CTL diet ($P < 0.001$). Total chewing time, chewing per DMI, NDF, and uNDF$_{48}$, were affected by period.

DISCUSSION

Diet Composition and Digestibility

In this study, we examined the use of forage uNDF$_{30}$ and forage uNDF$_{48}$ as factors in diet formulation to reduce the diet forage content and tested the effect on digestibility and performance. We determined the in vitro digestibility of the forages and diets with incubation times of 30, 48, and 240 h and calculated the uNDF. As expected, the uNDF at 30 and 48 h were higher in wheat straw than in wheat hay, and our digestibility values of these feedstuffs are within the range of those published in NASEM (2021). As can be seen, the uNDF$_{30,}$ uNDF$_{48}$, and uNDF$_{240}$ values were higher in the LF diet than in the CTL diet, which can be attributed to the high wheat straw content in the LF diet. The negative effect on digestibility of including straw in the milking-cow diet was demonstrated by Wang et al. (2014) for rice straw and Farmer et al. (2014) and Ben Meir et al. (2021) for wheat straw. The wheat straw rate in the diet in the study by Ben Meir et al. (2021) was similar to that in our study (7%–8%), and the decrease in NDF digestibility compared with the control diet was also similar (15%–16%). However, Kahyani et al. (2019a) found that including up to 22.3% of the diet with wheat straw, while balancing the diets for forage uNDF$_{30}$ and NDFd$_{30}$, increased the NDF digestibility. They explained that the higher digestibility in this diet was due to the inclusion of beet pulp and the greater rumen pH in the high-straw diet. However, in our study, pH <5.8 occurred at a longer time during the day, and the mean pH was lower for 9 h under the LF diet rather than the CTL diet, which might negatively affect the nutrient digestibility.

Particle Size and Feed Sorting

The particle distribution of the diets was similar in 3 out of the 4 sieves of the PSPS, and the fraction of particles >4 and <8 mm was higher in the CTL cows, which might explain the higher sorting for this fraction in cows fed the CTL diet at 10 h from feeding time. These results, indicating different feed sorting within the first hours

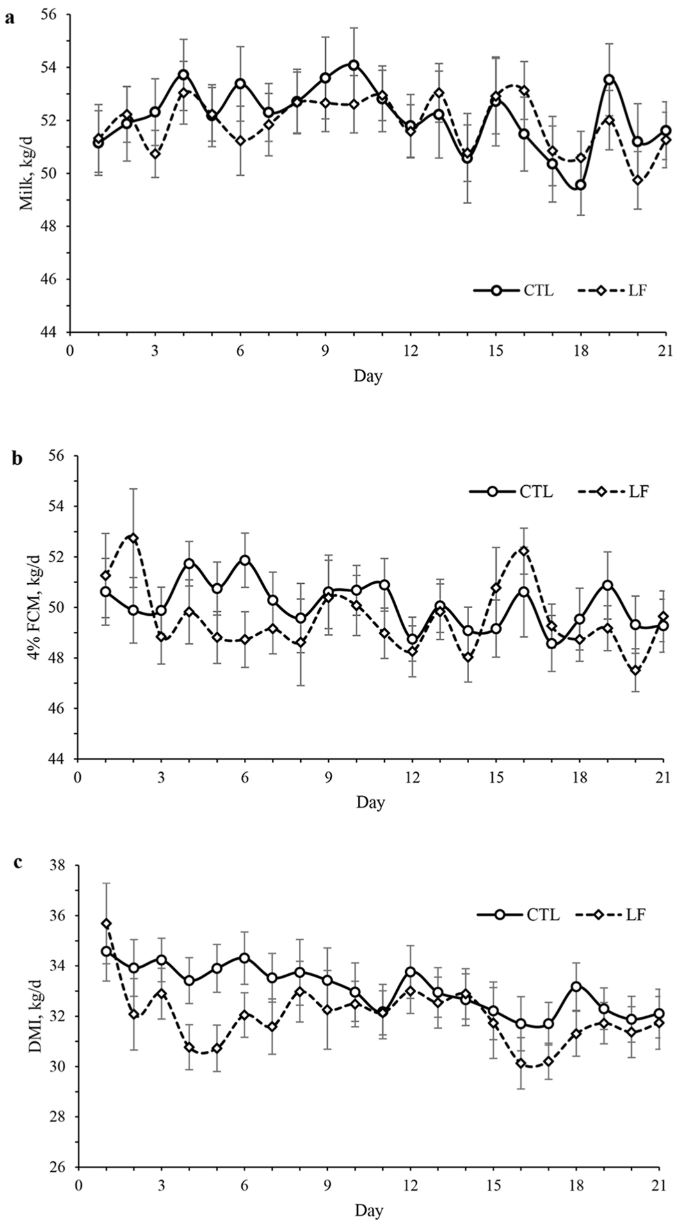

Figure 2. Average daily milk yields (a), 4% FCM yields (b), and DMI (c) of cows fed CTL or LF diets. Error bars represent the SEM.

after feed delivery, agree with the results of Maulfair et al. (2010), who examined the sorting index several times post-feeding. Importantly, Maulfair et al. (2010) found a higher difference in the sorting index between diets until 12 h post-feeding, which was diminished later, especially for the small particles. In our study, we did not record the sorting index after 24 h from feeding time. Particle size distribution of the diet might affect the feed intake, rumen pH, and production parameters (Zebeli et al., 2008); however, the differences in particle size distribution of the diets in the current study were negligible.

Production Performance and Efficiency

Although the milk yields were significantly (P = 0.009) higher in the CTL treatment, the differences were minor, with no differences in the milk fat and protein concentrations. The reduction in yields in cows fed the LF diet can be explained by the decline in the digestibility (9.3% for DM and 16.3% for NDF) of nutrients in this diet compared with the CTL diet. This was also found by Ben Meir et al. (2021), who replaced 8.2% corn grains with 7.5% wheat straw and 0.7% soybean meal and found a 3.5% reduction in DM digestibility in the latter diets, along with lower yields compared with the control cows. However, Kahyani et al. (2019a) fed cows a diet with 11.2% and 22.3% wheat straw, which were balanced for forage $uNDF_{30}$ and $NDFd_{30}$, and reported a substantial increase in the digestibility of all nutrients in the high-straw diet, which was not reflected in the milk yields in these groups. Li et al. (2020) fed cows diets containing a 35:65 and 60:40 forage-to-concentrate ratio and found a decrease in digestibility in the latter diet, but with higher milk and ECM yields. The extent of fiber digestibility depends on the size of the indigestible fraction, and the competition between the rate of ruminal passage and degradation (Oba and Allen, 1999), which is influenced by the intake level. According to Gabel et al. (1992), the digestibility of DM and all specific measures of dietary components decline significantly as the nutrition level increases, which means that DMI also increases. Among many other factors, the yield outcomes are influenced by nutrient digestibility and feed intake, which consequently determine the total digested nutrients.

In this study, the inclusion of wheat straw instead of wheat hay decreased the feed intake, digestibility, and yields; however, the efficiency of milk production was enhanced, but not for FCM and ECM, because of the trend of lower fat and protein content in the milk in the LF than CTL treatment. In Kahyani et al. (2019a), the efficiency of ECM production was linearly higher in cows fed diets with increased wheat straw content that were balanced for forage $uNDF_{30}$ and $NDFd_{30}$. Li et al. (2020) reported higher efficiency in cows fed with the 60:40 forage-to-concentrate ratio in diets than with the 35:65 ratio. It is well established that increasing the forage content in the diet, especially straw, decreases the intake and improves production efficiency (Ferris et al., 2000; Eastridge et al., 2017; Ben Meir et al., 2018; Kahyani et al., 2019a; Hanlon et al., 2020). However, in our study, the LF diet was 5.2% lower in forage content, but the forage $uNDF_{30}$ and $uNDF_{48}$ fractions were similar to those in the CTL diet, and we observed a similar trend of lower intake and higher efficiency.

In the current study, the effect of period was significant regarding several parameters: rumen pH, total-tract

Table 7. Least squares means of intake, BW, and intake per metabolic BW ($BW^{0.75}$)

Parameter[2]	Treatment[1] CTL	LF	SEM	*P*-value Treatment	Period
Intake, kg/d					
DM	32.7	31.7	0.84	<0.001	0.003
NDF	11.3	11.6	0.29	0.41	0.12
$uNDF_{30}$	7.1	7.4	0.19	<0.001	<0.001
$uNDF_{48}$	5.6	6.2	0.15	0.002	0.12
$uNDF_{240}$	3.4	4.0	0.09	<0.001	0.12
Forage $uNDF_{30}$	3.9	3.7	0.10	<0.001	<0.001
Forage $uNDF_{48}$	3.3	3.3	0.08	0.74	0.12
Forage $uNDF_{240}$	2.1	1.8	0.05	0.001	0.13
$BW^{0.75}$, kg	131.1	130.5	1.33	0.76	0.81
Intake/$BW^{0.75}$, g/d					
DM	249.4	242.7	6.07	0.43	0.09
NDF	85.6	89.4	2.25	0.23	0.06
$uNDF_{30}$	53.9	56.9	1.42	<0.001	<0.001
$uNDF_{48}$	42.4	47.7	1.19	0.003	0.17
$uNDF_{240}$	27.9	30.3	0.80	0.04	0.12
Forage $uNDF_{30}$	29.9	28.8	0.73	<0.001	<0.001
Forage $uNDF_{48}$	25.0	25.9	0.85	0.47	0.09
Forage $uNDF_{240}$	17.9	13.5	0.72	<0.001	0.002

[1]CTL = 35.7% forage (DM basis) based on 20% wheat silage, 15.8% wheat hay, 11.8% forage $uNDF_{30}$, and 10.0% forage $uNDF_{48}$ or LF = 30.5% forage based on 20% wheat silage, 2.2% wheat hay, 8.3% wheat straw 11.8% forage $uNDF_{30}$, and 10.4% forage $uNDF_{48}$.

[2]NDF = NDF measured using amylase and sodium sulfite; $uNDF_{30}$ = uNDF estimated through 30 h of in vitro incubation; $uNDF_{48}$ = uNDF estimated through 48 h of in vitro incubation; $uNDF_{240}$ = uNDF estimated through 240 h of in vitro incubation.

in vivo digestibility, DMI, and rumination time. All these changes may be related to the increased THI during the second period, and the well-known effects of heat stress on the cow's metabolism (Sammad et al., 2020). However, we did not find any significant interaction between treatment and period, suggesting that these dietary changes probably had no effect on heat production, in contrast to those found by Adin et al. (2008), who replaced wheat silage with soybean hulls.

Feed Intake and Recumbence, Rumination, Eating, and Panting Times

In our study, a higher intake of $uNDF_{30}$, $uNDF_{48}$, and $uNDF_{240}$ was found in the LF than in the CTL diet, which was accompanied by lower DMI in this treatment. Kahyani et al. (2019a) fed cows different wheat straw contents and found a similar $uNDF_{30}$ intake among the groups, along with a tendency of differences in feed intake between groups, which is in contrast with our findings for $uNDF_{30}$ and $uNDF_{48}$ intake. Fustini et al. (2017) fed cows diets with different levels of alfalfa hays differing in NDF digestibility and found no differences in $uNDF_{24}$ intake but with considerable differences in the total DMI. In the current study, the intake of forage $uNDF_{30}$ and forage $uNDF_{30}$ per $BW^{0.75}$, or forage $uNDF_{240}$ and forage $uNDF_{240}$ per $BW^{0.75}$ were higher in the CTL than in the LF treatment. However, despite the lower DMI in the LF diet, we found no differences in the forage $uNDF_{48}$ intake or forage $uNDF_{48}$ intake per $BW^{0.75}$ (but not for forage $uNDF_{30}$ intake) between treatments, which suggests that under our study conditions, forage $uNDF_{48}$ played a substantial role in influencing the feed intake. In Kahyani et al. (2019a), the diets were balanced for forage $uNDF_{30}$ as in our study, but the forage $uNDF_{30}$ intake was not reported. Kahyani et al. (2019a) explained their results by the high percentage of beet pulp soluble fiber in their diet, which provided more energy and improved digestibility.

We calculated the intake of several parameters per the metabolic BW, and except for the DM, all other results had a trend similar to the raw intake results. The eating time for most of the variables was higher for the LF diet than in the CTL diet, which might be explained by the high wheat straw content in this diet, which was also found in the study by Ben Meir et al. (2021). According to Kahyani et al. (2022), the eating time for diets that were balanced for forage $uNDF_{30}$ and $NDFd_{30}$ and did not include wheat straw was higher than for those diets with up to 22.3% wheat straw. However, in the current study, the LF diet included 8.3% wheat straw, and the eating time was higher compared with the CTL diet. Furthermore, in Kahyani et al. (2022), there was an 18-percentage-unit difference in the total forage between diets, which might influence the total eating time. Interestingly, in contrast to the eating time, the rumination time was higher for

Table 8. Daily time budgets of cows in the study

Parameter[2]	Treatment[1] CTL	LF	SEM	*P*-value Treatment	Period
Resting time, min/d	520	524	10.5	0.34	<0.001
Panting, min/d	140	145	18.6	0.38	<0.001
Eating					
Min/d	203	213	1.9	<0.001	<0.001
Min/kg of DM	6.3	6.9	0.30	<0.001	0.02
Min/kg of NDF	17.8	18.7	0.82	<0.001	<0.001
Min/kg of $uNDF_{30}$	28.6	29.2	1.31	0.16	<0.001
Min/kg of $uNDF_{48}$	35.1	34.8	1.57	0.42	<0.001
Min/kg of $uNDF_{240}$	56.3	54.4	2.49	<0.01	<0.001
Min/kg of forage $uNDF_{30}$	51.5	57.7	2.50	<0.001	<0.001
Min/kg of forage $uNDF_{48}$	61.7	65.9	2.88	<0.001	0.001
Min/kg of forage $uNDF_{240}$	106.2	108.8	4.83	0.05	<0.001
Rumination					
Min/d	534	522	8.7	0.001	0.03
Min/kg of DM	16.6	16.9	0.47	0.08	<0.001
Min/kg of NDF	47.2	46.1	1.31	0.03	0.008
Min/kg of $uNDF_{30}$	77.9	72.3	2.07	<0.001	0.003
Min/kg of $uNDF_{48}$	93.5	85.9	2.55	<0.001	0.88
Min/kg of $uNDF_{240}$	149.9	134.4	4.08	<0.001	0.40
Min/kg of forage $uNDF_{30}$	140.3	143.0	4.03	0.18	<0.001
Min/kg of forage $uNDF_{48}$	163.9	162.7	4.56	0.48	0.001
Min/kg of forage $uNDF_{240}$	282.5	268.6	7.75	<0.001	0.13
Total chewing					
Min/d	737	734	12.6	0.55	<0.001
Min/kg of DM	22.8	23.7	0.64	<0.001	0.008
Min/kg of NDF	65.0	64.7	1.75	0.71	0.47
Min/kg of $uNDF_{30}$	106.5	101.4	2.79	0.25	<0.001
Min/kg of $uNDF_{48}$	128.7	120.7	3.36	<0.001	0.05
Min/kg of $uNDF_{240}$	206.3	188.7	5.33	<0.001	0.002
Min/kg of Forage $uNDF_{30}$	191.9	200.6	5.41	0.22	<0.001
Min/kg of Forage $uNDF_{48}$	225.7	228.4	6.19	0.24	0.15
Min/kg of Forage $uNDF_{240}$	388.9	377.2	10.36	0.003	0.61

[1]CTL = 35.7% forage (DM basis) based on 20% wheat silage, 15.8% wheat hay, 11.8% forage $uNDF_{30}$, and 10.0% forage $uNDF_{48}$ or LF = 30.5% forage based on 20% wheat silage, 2.2% wheat hay, 8.3% wheat straw 11.8% forage $uNDF_{30}$, and 10.4% forage $uNDF_{48}$.

[2]NDF = NDF measured using amylase and sodium sulfite; $uNDF_{30}$ = uNDF estimated through 48 h of in vitro incubation; $uNDF_{48}$ = uNDF estimated through 240 h of in vitro incubation; $uNDF_{240}$ = uNDF estimated through 240 h of in vitro incubation.

cows fed the CTL diet. We can speculate that the longer eating time is also accompanied by a higher chewing time per kilogram of DMI in this group, which might help to decrease the fiber length and consequently reduce the rumination time. However, the rumination and chewing times per kilogram of forage $uNDF_{30}$ or forage $uNDF_{48}$ did not differ between diets.

In the current study, we balanced the diets for forage $uNDF_{30}$ and forage $uNDF_{48}$ and found similar trends in most parameters that were examined, except forage $uNDF_{30}$ intake, which was higher in the CTL than in the LF diet but was similar for the forage $uNDF_{48}$. However, Lopes et al. (2015) compared the in vitro 30- and 48-h NDF digestibility values with in vivo total-tract NDF digestibility and found a better relationship with 48 h than 30 h. Consequently, NASEM (2021) adopted the 48-h incubation as more accurate for estimating the in vivo digestibility by lactating cows.

In the current study, we did not balance the diets for NDFd, but for forage uNDF (after 30 and 48 h of in vitro incubation). We can assume that balancing the diets for NDFd, as was done in Kahyani et al. (2019a,b), may influence the outcomes, mainly because it takes into account the total NDF in the diet. Another limitation of this study is the relatively high proportion of particle sizes >19 mm (~20%) in the diets, which might influence the feed sorting and intake, digestibility, and cows' performance.

CONCLUSIONS

In this study, we fed high-yielding cows a CTL diet containing 35.8% forage or an LF diet containing 30.6% forage; diets were balanced for forage $uNDF_{30}$ and forage $uNDF_{48}$ by replacing wheat hay with wheat straw. The DMI and the digestibility of DM and most nutrients were reduced in the LF diet, which can be attributed to the high wheat straw content in this diet compared with the CTL diet. The milk, 4% FCM, and ECM yields were higher in the CTL than in the LF treatment, but the milk-to-DMI ratio was higher in the LF than in the CTL treatment. It can be concluded that balancing the diet for forage uNDF$_{30}$ or forage uNDF$_{48}$ might be considered as an approach to reduce the forage content in the diet; however, further research is required to precisely calculate the diet composition to avoid a reduction in digestibility and yields.

NOTES

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Nonstandard abbreviations used: BW$^{0.75}$ = metabolic BW; CNCPS = Cornell Net Carbohydrate and Protein System; CTL = control; DDGs = dried distillers grains with solubles; EE = ether extract; iNDF = indigestible NDF; LF = low-forage diet; NDFd = NDF digestibility; NDFd$_{30}$ = NDFd estimated through 30 h in vitro incubation; paNDF = physically adjusted NDF; peNDF = physical effective NDF; PSPS = Penn State Particle Separator; THI = temperature-humidity index; uNDF = undigested NDF; uNDF$_{24}$ = uNDF estimated through 24 h of in vitro incubation; uNDF$_{30}$ = uNDF estimated through 30 h of in vitro incubation; uNDF$_{48}$ = uNDF estimated through 48 h of in vitro incubation; uNDF$_{240}$ = uNDF measured at 240 h of in vitro incubation.

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